



10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married		
	Yes	No	
Bachelors or higher	187 66	113 34	300
Some college	162 66	138 34	300
High school or less	149 66	151 34	300
	498	402	900

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

10.

H_0 : marry rate and educational level are independent

H_a : are dependence

類別 vs 類別 $\rightarrow \chi^2$ 獨立

$$E = \frac{300 \times 498}{900} = 166$$

$$E = \frac{300 \times 402}{900} = 134$$

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(187-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(162-166)^2}{166} + \dots + \frac{(151-134)^2}{134} = 10.0611$$

$$\chi^2_{(2-1)(3-1), 0.01} = \chi^2_{2, 0.01} = 9.21$$

$\chi^2 > \chi^2_{2, 0.01}$ = Reject $H_0 \rightarrow$ 有足夠 evidence 顯示 marriage rate 與 educational level 有顯著相關

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence			
	Apartment	Duplex	Single Residence	
1	8 9.67	20 9.67	1 9.67	29
2	16 11	8 11	9 11	33
3	10 11.33	10 11.33	14 11.33	34
4 or More	6 8	2 8	16 8	24
	40	40	40	120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

12.

H_0 : Family size of three types of residence distribution are identical

H_a : are not identical

類別 vs 類別 (分佈) $\rightarrow$ 卡方獨立

$$E = \frac{29 \times 40}{120} = 9.67 \dots \text{每個計算}$$

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(8-9.67)^2}{9.67} + \dots + \frac{(16-8)^2}{8} = 36.499 \text{ 查表夠大?}$$

$$\chi^2_{(k-1)(b-1), 0.01} = \chi^2_{6, 0.01} = 16.81$$

$\chi^2 > \chi^2_{6, 0.01}$ = Reject $H_0 \rightarrow$ Family size of three types of residence are significant differ

+ 看表說故事:

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	2.5	10	7.5	5	9
	4.8	5.2	5.0	4.5	5.1
Below Town	5.0	4.7	4.9	4.8	4.9
	4.6		2.5		

- Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.
- Use a Student's t -test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

b) t -test

$$H_0: \mu_1 = \mu_2$$

$$H_a: \mu_1 < \mu_2$$

$$t = \frac{\bar{X}_A - \bar{X}_B - 0}{\sqrt{s^2(\frac{1}{n} + \frac{1}{n})}} = \frac{5 - 4.86}{\sqrt{0.11(\frac{1}{5} + \frac{1}{5})}} = 1.606$$

$$\text{查表 } t_{10-2, 0.05} = 1.86$$

$$t^* < t_{0.05} \rightarrow \text{Non-reject } H_0 \text{ 同 (a)}$$

12.

a)

H_0 : Above and Below 含氧量分布相同

H_a : Below > Above

Wilcoxon rank sum

$$T_1 = 2.5 + 10 + 7.5 + 5 + 9 = 34 \quad \text{Total } \frac{n(n+1)}{2} = \frac{10 \times 11}{2} = 55$$

$$T_2 = 21$$

$$T_1^* = n_1(n_1 + n_2 + 1) - T_1 = 5 \times (5 + 5 + 1) - 34 = 21$$

$$\min(T_1, T_1^*) = (34, 21) = 21 \text{ 是否够小?}$$

查表 $n_1 = 5, n_2 = 5$ 单尾 $\alpha = 0.05$

$\Rightarrow$ critical point = 19 < 21 $\Rightarrow$ Non-reject H_0

$\Rightarrow$ 没有足够 evidence 显示污染会降低含氧量

DATA SET
D51502

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

Deaf Children	Hearing Children
2.75 (15)	.89 (1)
2.14 (11)	1.43 (7)
3.23 (18)	1.06 (4)
2.07 (10)	1.01 (3)
2.49 (14)	.94 (2)
2.18 (12)	1.79 (8)
3.16 (17)	1.12 (5.5)
2.93 (16)	2.01 (9)
2.20 (13)	1.12 (5.5)
Rank Sum	126
	45

13.

眼睛

H_0 : 正常與失聰 转移率分布相同

H_a : 分布不完全相同

$$T_1 = 126, T_2 = 45 \quad \text{Total} = \frac{18(18+1)}{2} = 9 \times 19 = 171$$

$$T_1^* = n_1(n_1 + n_2 + 1) - T_1 = 9(9 + 9 + 1) - 126 = 45$$

$$\min(T_1, T_1^*) = 45 \Rightarrow \text{查表 } n_1 = 9, n_2 = 9, \text{ 雙尾 } \alpha = 0.05 = 62$$

$45 < 62 \Rightarrow$ Reject $H_0 \Rightarrow$ 失聰與正常兒童 eye-movement rate 有顯著不同

Two Simple Examples Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \leq .10$. Calculate the observed value of the test statistic and present your conclusion.

4.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	+	+	+	-	+	+	+

$$\Rightarrow X = 6$$

4.

H_0 : 兩母體分佈相同

H_a : 分佈不完全相同

$$X = 6, \alpha = 0.1, n = 7$$

$$X \sim \text{Bin}(7, 0.5)$$

$$P(X \geq 6) \text{ or } P(X \leq 1)$$

$$= C_0^7 0.5^7 + C_1^7 0.5^7 + C_6^7 0.5^7 + C_7^7 0.5^7$$

$$= 0.078125 + 0.546875 = 0.625 > 0.1$$

$$P(X > 6) \text{ or } P(X < 1)$$

$$= C_0^7 0.5^7 + C_1^7 0.5^7 = 0.078125 + 0.546875$$

$$= 0.625 < 0.1$$

P.P. $X = 0, X = 7 \Rightarrow X = 6 \notin R.R.$
 $\Rightarrow$ Non-reject H_0

DATA SET DS1507 11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1–10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal	A	B	Meal	A	B
1	6	8	12	8	5
2	4	5	13	4	2
3	7	4	14	3	3
4	8	7	15	6	8
5	2	3	16	9	10
6	7	4	17	9	8
7	9	9	18	4	6
8	7	8	19	4	3
9	2	5	20	5	4
10	4	3	21	3	2
11	6	9	22	5	3

$$t = 11$$

$$- = 9$$

- Use the binomial tables in Appendix I to find the exact rejection region for the test.
- Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)
- Compare the results of parts a and b.

b)

$$Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{11 - 11}{\sqrt{11(1-1)}} = 0.047 < Z_{0.05} = 1.96$$

$$\Rightarrow \text{Non-reject}$$

11.

$$n = 20, X = 11$$

a)

$$X \sim \text{Bin}(20, 0.5)$$

$$\alpha = 0.05, \text{ 雙尾 } \Rightarrow \alpha = 0.025$$

$$P(X \leq 1) = C_0^{20} 0.5^{20} + C_1^{20} 0.5^{20}$$

$$P(X \leq 2) = P(X \leq 1) + C_2^{20} 0.5^{20}$$

$$P(X \leq 3) = P(X \leq 2) + C_3^{20} 0.5^{20}$$

$$P(X \leq 4) = P(X \leq 3) + C_4^{20} 0.5^{20} = 0.0007$$

$$\approx 0.001$$

R.R. = $X \leq 4$ or $X \geq 15$, $X = 11 \notin R.R.$
 $\Rightarrow$ Non-reject H_0

13. **Recovery Rates** Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

- a. Test using the sign test. Choose your rejection region so that α is near .05.
- b. Why might it be inappropriate to use the Student's t -test in analyzing the data?

Hospital	Drug A		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

Hospital	Drug B		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

13.

a) Sign test

$$X = 3, \quad n = 10$$

$$P(X=1) = C_0^{10} 0.5^{10} + C_1^{10} 0.5^{10} = 0.0108$$

$$P(X=2) = 0.0108 + C_2^{10} 0.5^{10} = 0.0108 + 0.0439 = 0.0547$$

$$\Rightarrow (P(X=1) + P(X=2)) > 0.05 \approx 0.0547$$

$$R.D. \quad X \leq 1 \text{ or } X \geq 9. \quad X=3 \notin R.D.$$

$\Rightarrow$ Non-reject H_0

$$b) \quad t\text{-test} = \frac{\bar{X}_1 - \bar{X}_2 - 0}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$